



Session 8

Studies on schooling fish using sonar

Interpretation of acoustic data from fisheries sonar



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ICES Journal of Marine Science



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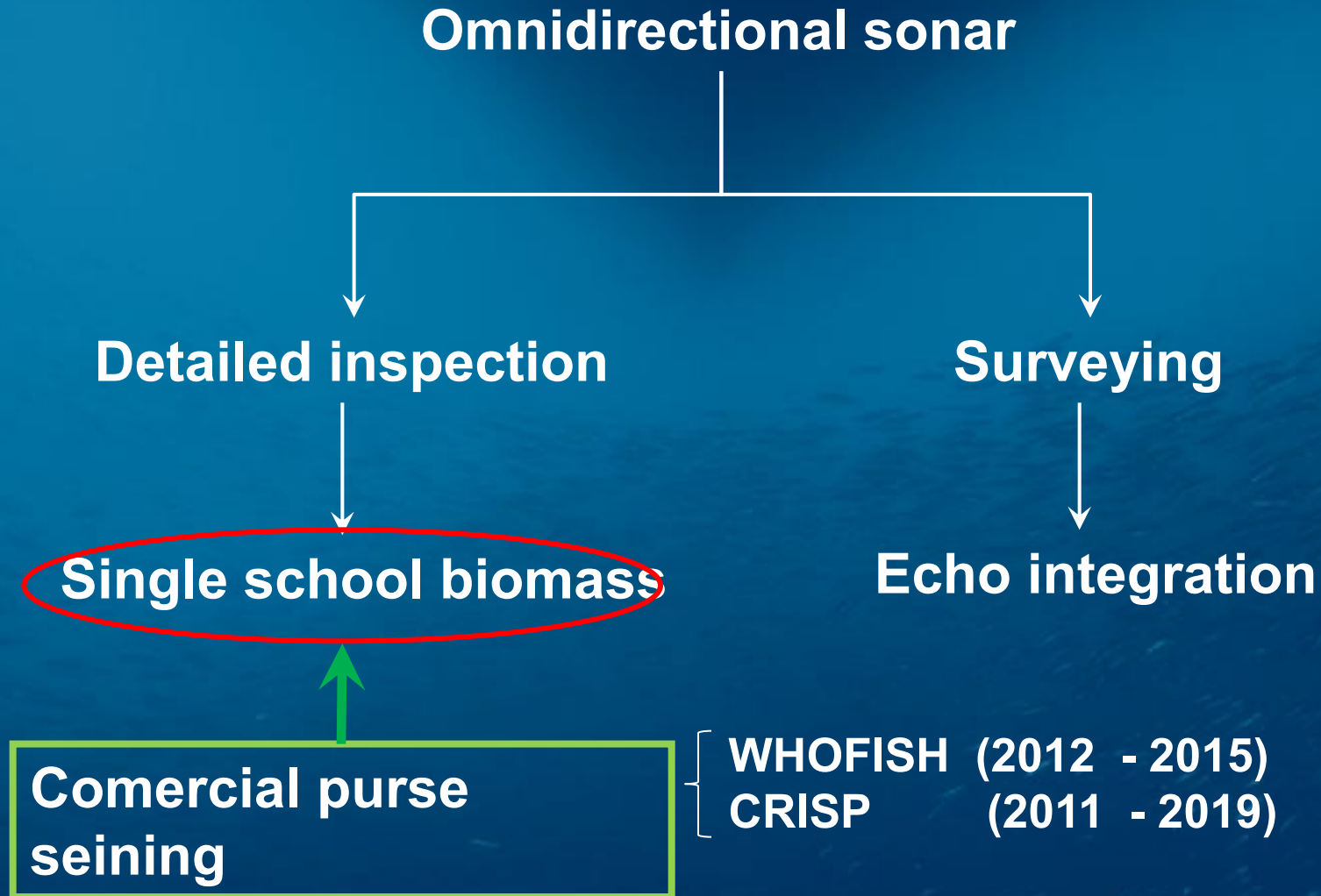
ICES Journal of Marine Science (2021), 78(3), 940 – 951. doi:10.1093/icesjms/fsaa237

Estimating individual fish school biomass using digital omnidirectional sonars, applied to mackerel and herring

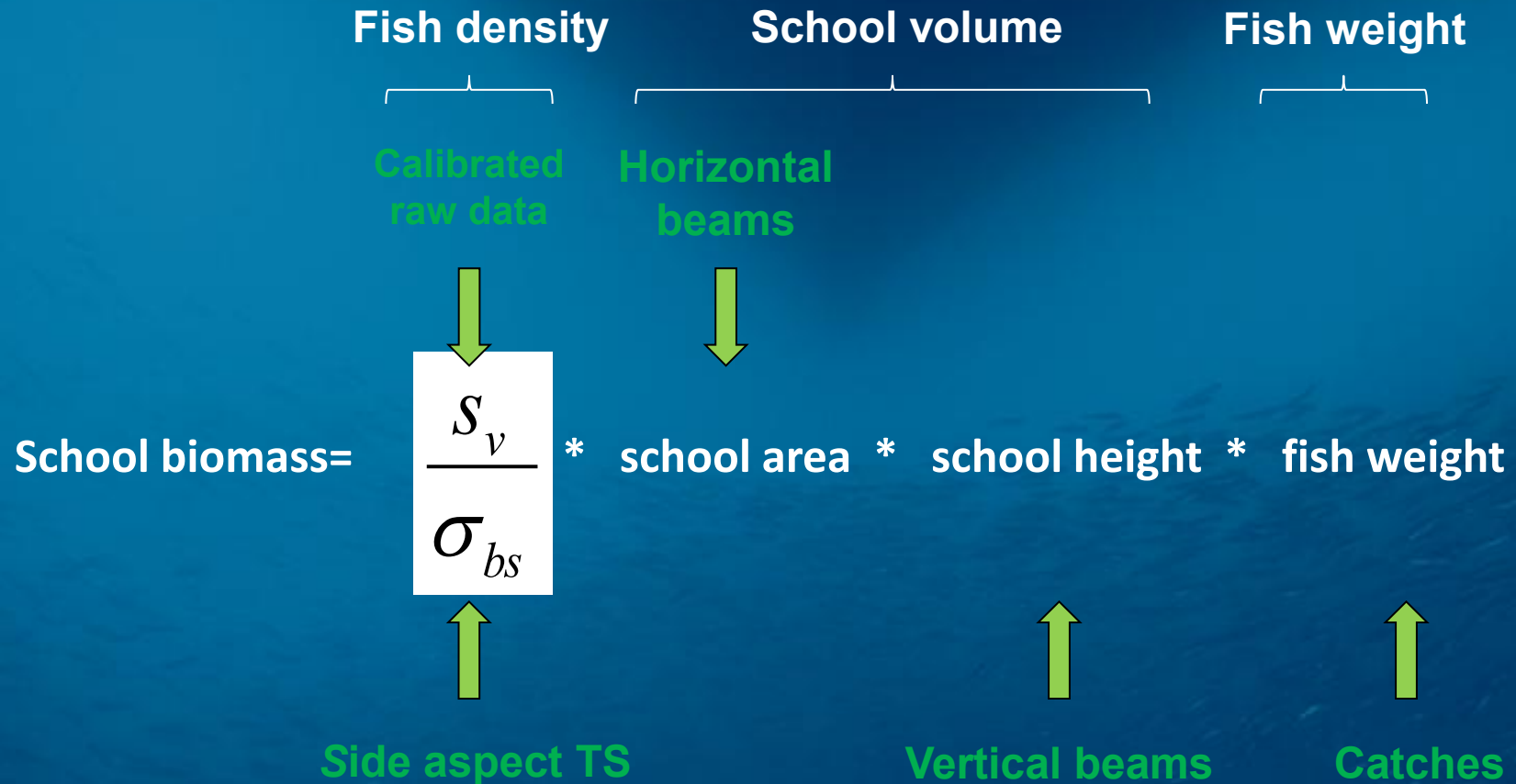
Héctor Peña  *, Gavin J. Macaulay  , Egil Ona, Sindre Vatnehol, and Arne J. Holmin



Motivation



Biomass calculation



Simrad SX90 and SU90 at 26 and 30 kHz



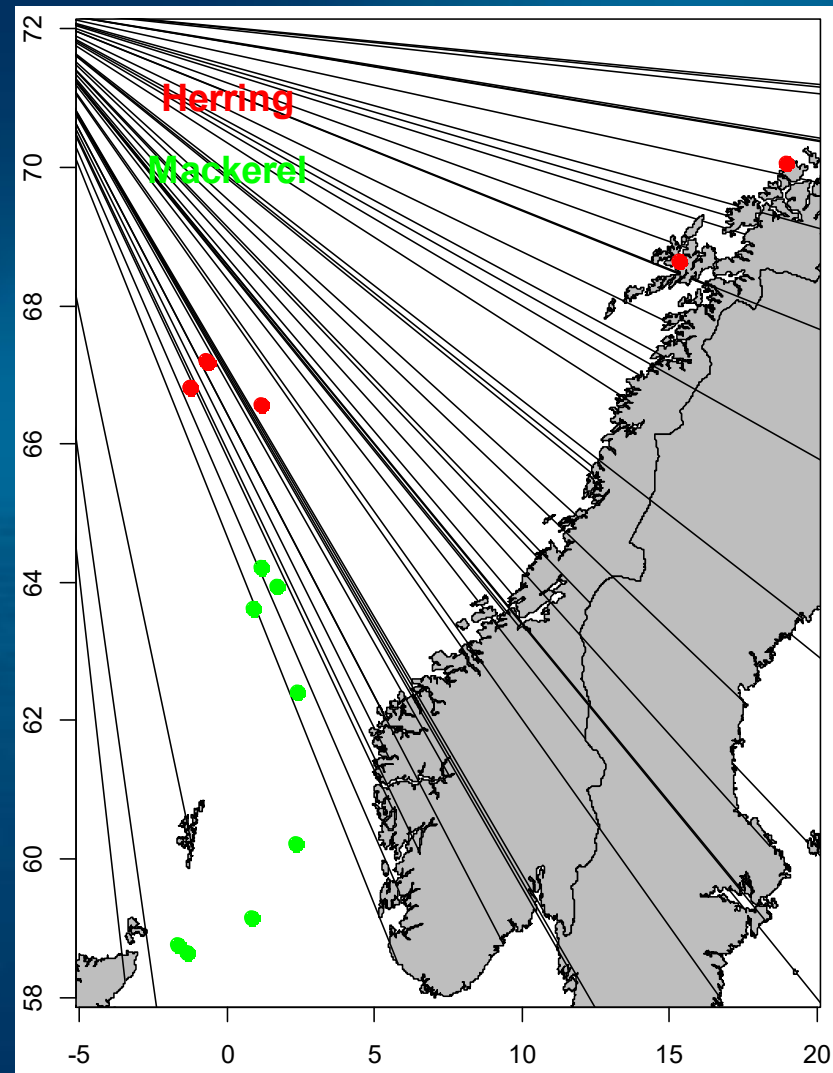
Results

Vessel	Dates	Ship days	Species	Schools
GO Sars	Nov. 2012	14	Herring	20
GO Sars Artus	Mar. 2013	28	Herring	1
GO Sars Artus	Nov. 2013	28	Herring	7
GO Sars Kings Bay	Oct. 2014	28	Mackerel	6
GO Sars Eros	Oct. 2015	28	Mackerel	5
GO Sars Brennholm	Oct. 2016	28	Mackerel	2
GO Sars Eros	Oct. 2017	30	Herring	5

Total of 184 ship days
46 best quality schools
15 verified catches

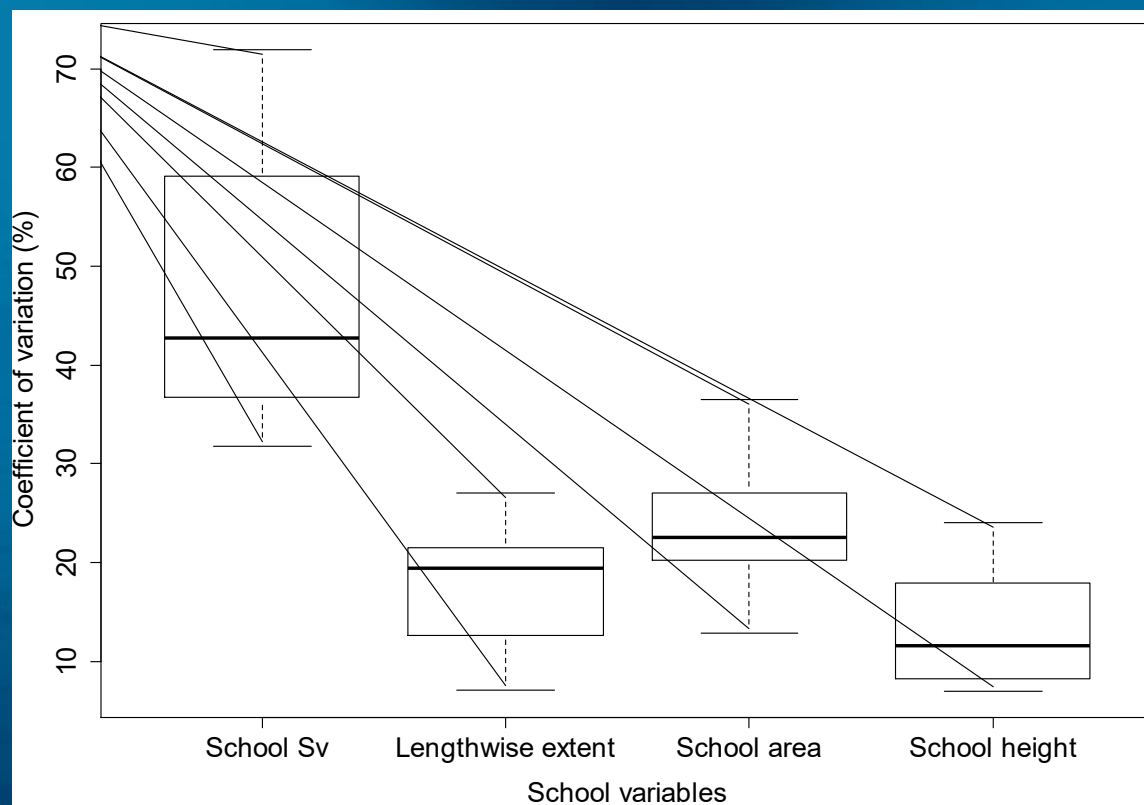


Schools processed



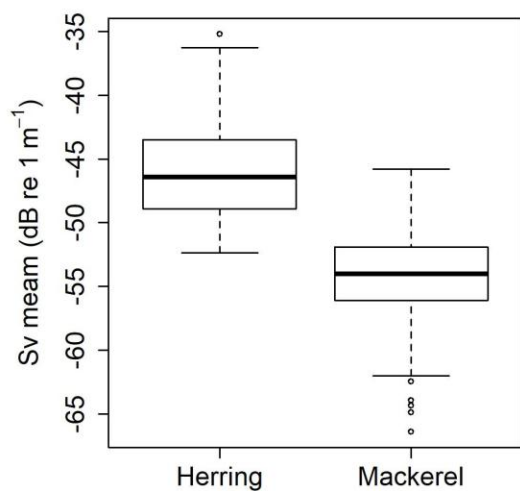
Results

Coefficient of variation for school parameters

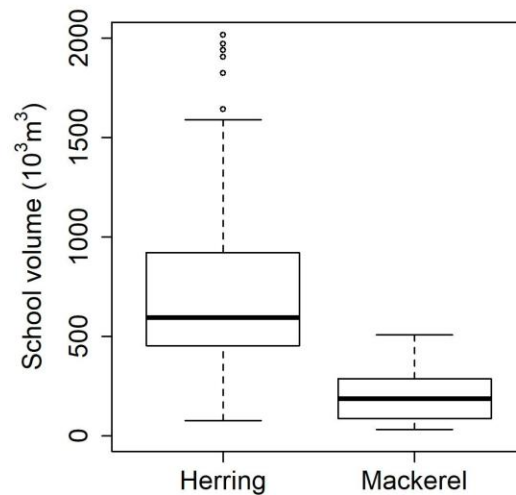


Results

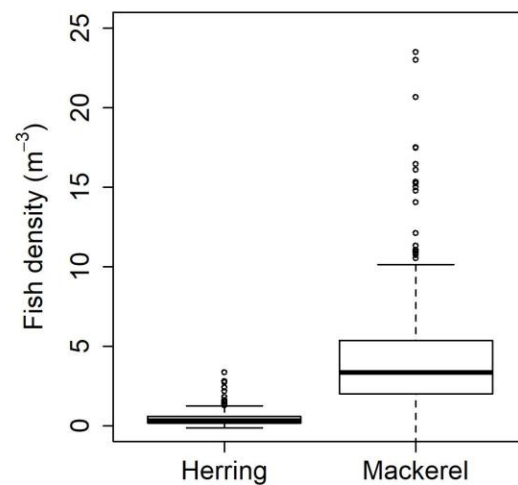
School volume backscattering strength



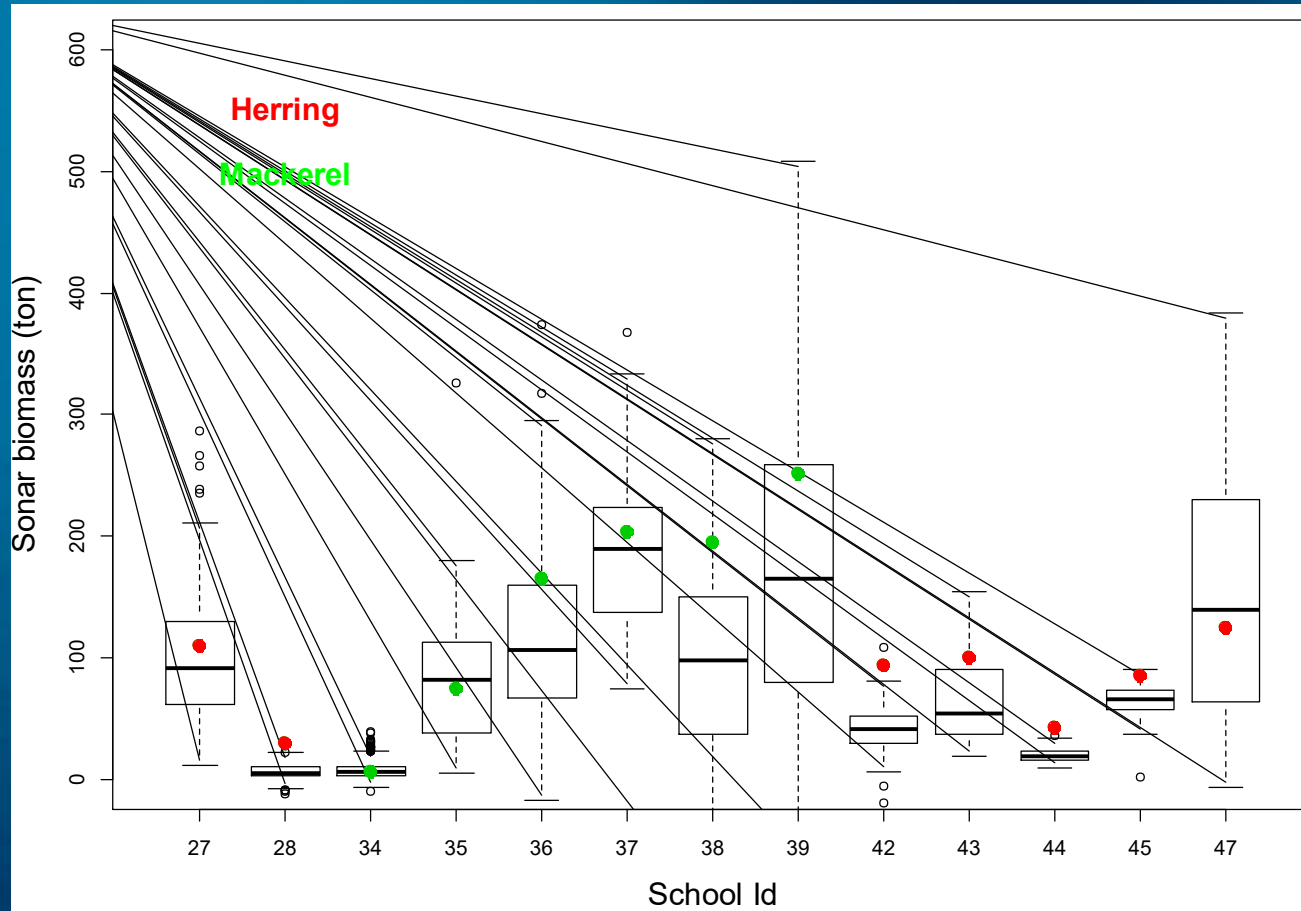
School volume



Fish density



School biomass estimate and real purse seine catches

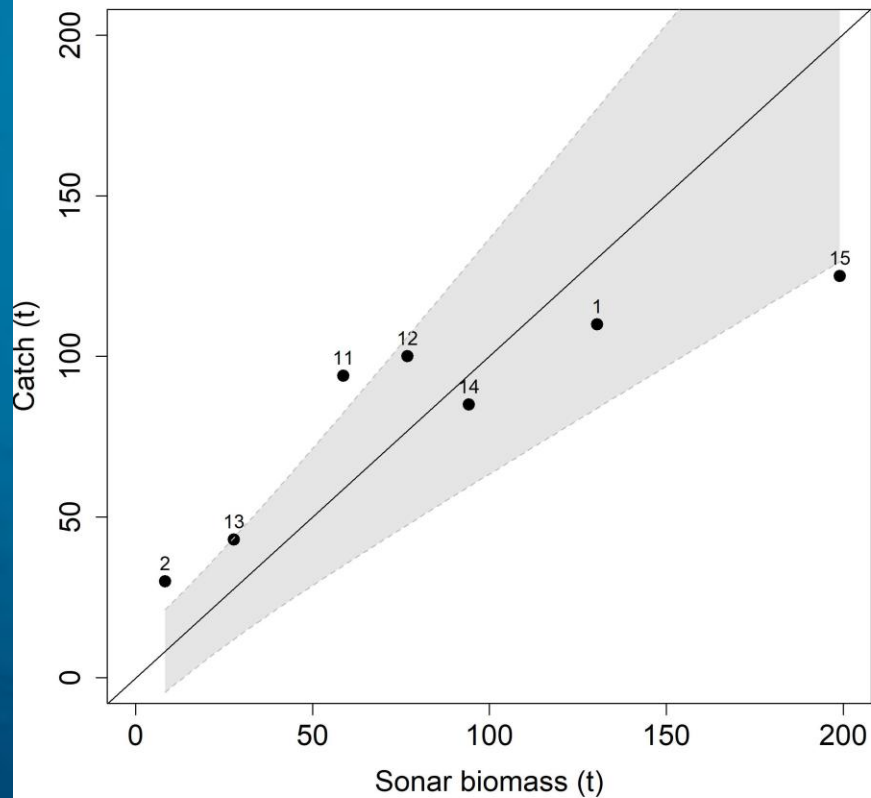


Id	Catch	Median sonar
27	110	130
28	30	8
34	6	8
35	75	106
36	165	138
37	203	245
38	195	127
39	251	215
42	94	59
43	100	77
44	43	28
45	85	94
47	125	199

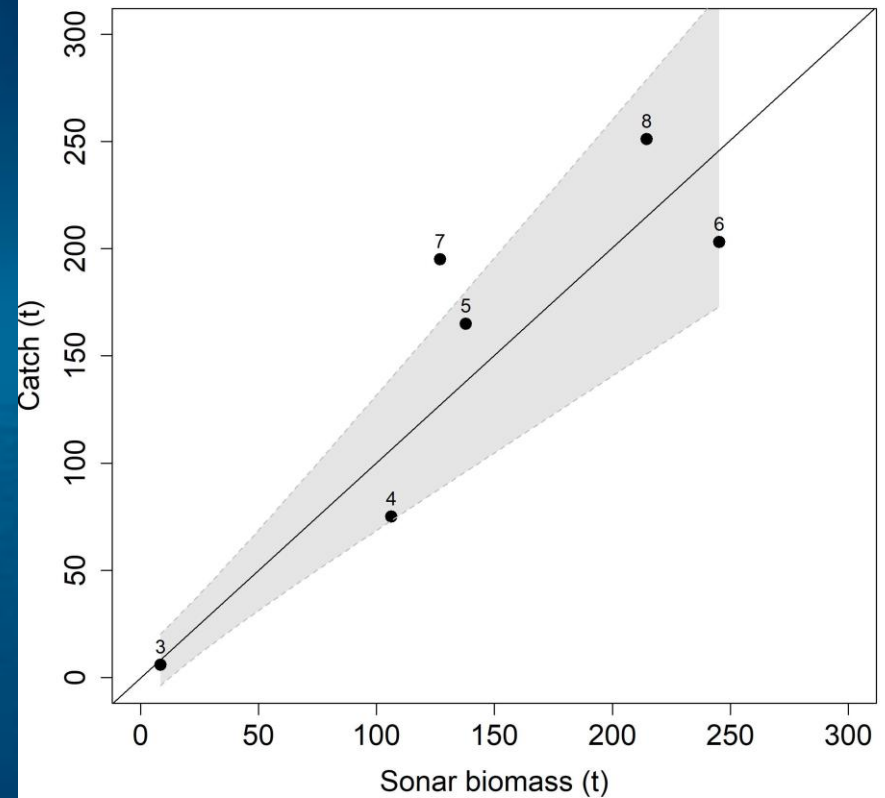


School biomass estimate and real purse seine catches

Herring



Mackerel



$$TS = 20 \log(L) - 2.3 \log\left(1 + \frac{z}{10}\right) - 65 + Corr$$

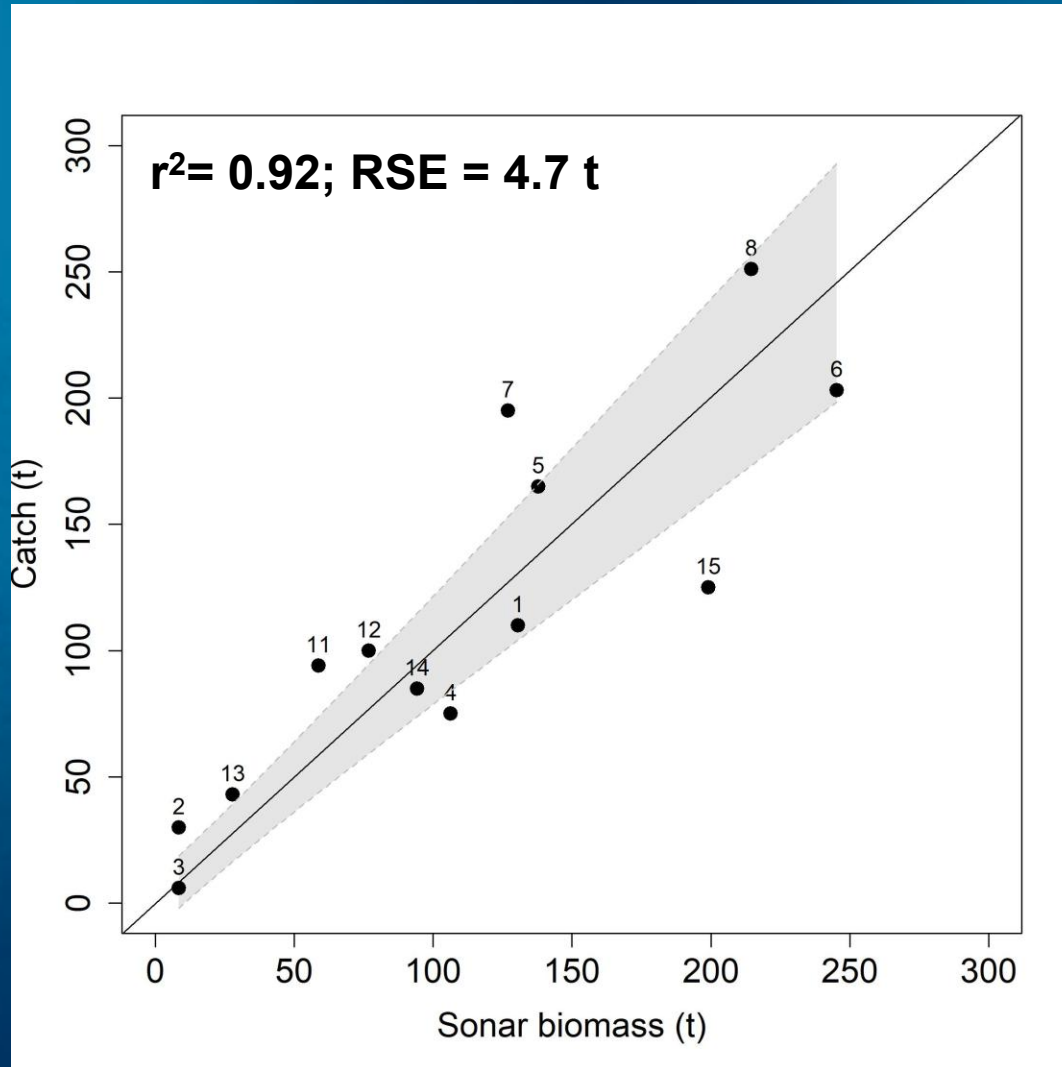
- 4.5 dB

$$TS = 20 \log(L) - 86 + Corr$$

- 4.1 dB

School biomass estimate and real purse seine catches

Combined mackerel and herring



School biomass estimate and real purse seine catches

R/V "GO Sars"



F/V "Kings Bay"



F/V "Eros"



F/V "Artus"



F/V "Brennholm"



School behaviour

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Feeding herring schools do not react to seismic air gun surveys

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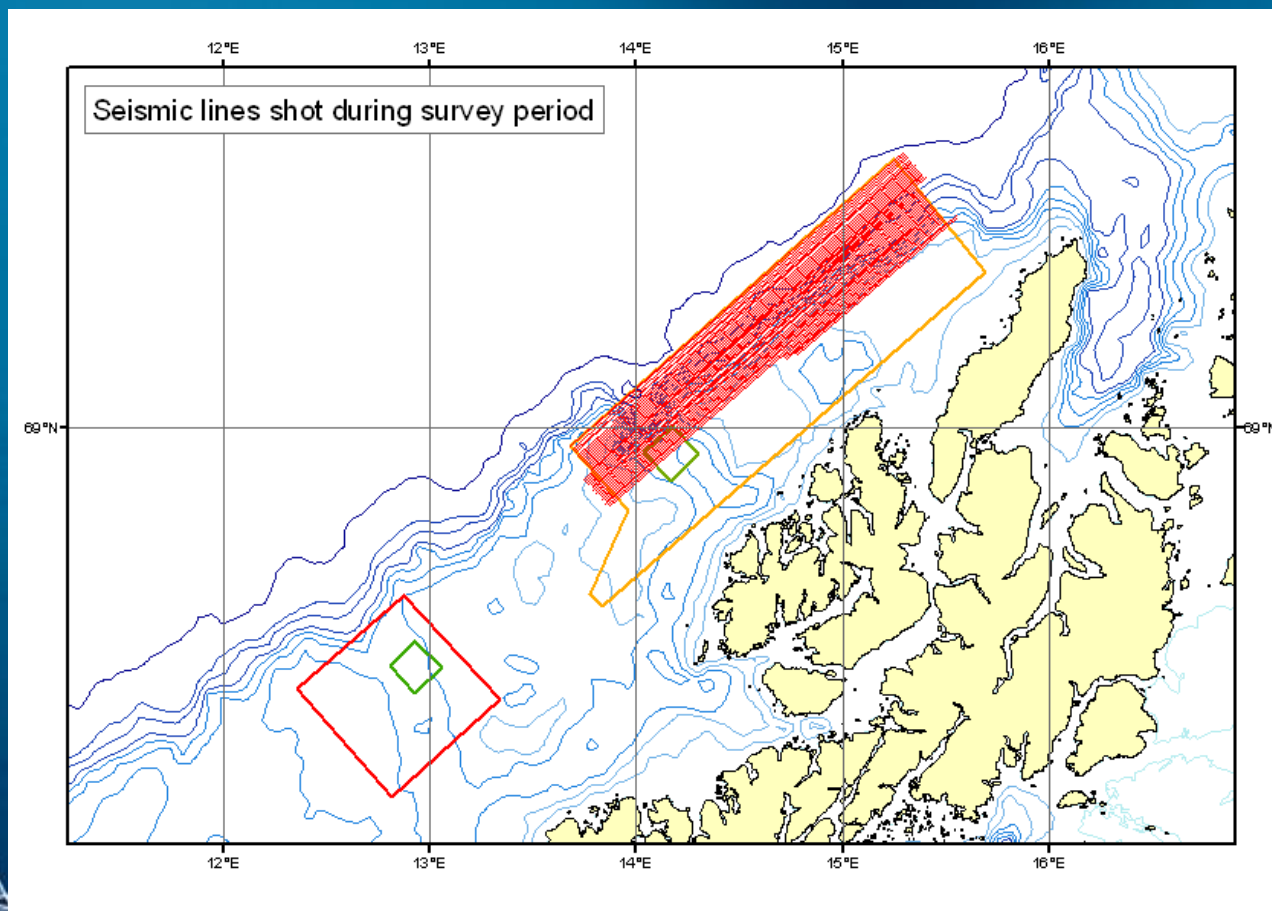
Peña, H., Handegard, N. O. and Ona, E. Feeding herring schools do not react to seismic air gun surveys. – ICES Journal of Marine Science, doi:10.1093/icesjms/fst079.

Received 23 January 2013; accepted 2 May 2013.



Study background

Effect of seismic surveys on fish distribution and catch rates of gillnets and longlines in Vesterålen in summer 2009



Seismic shooting
Start 29.06.09 09:51
End 06.08.09 06:46

Experimental set -up

Date: 29 June 2009

Start: 09:42 UTC

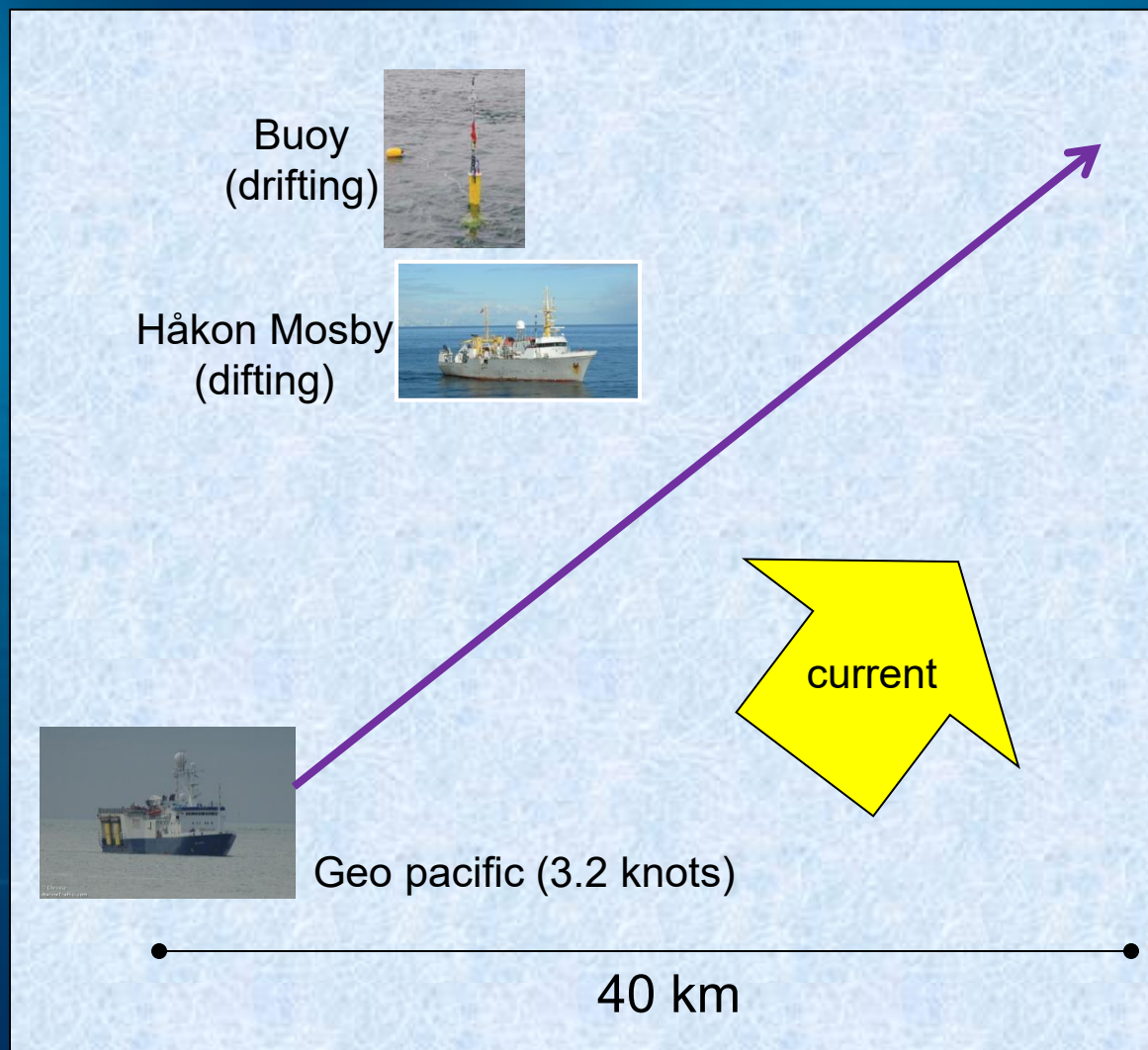
Finish: 15:02 UTC

Geo pacific

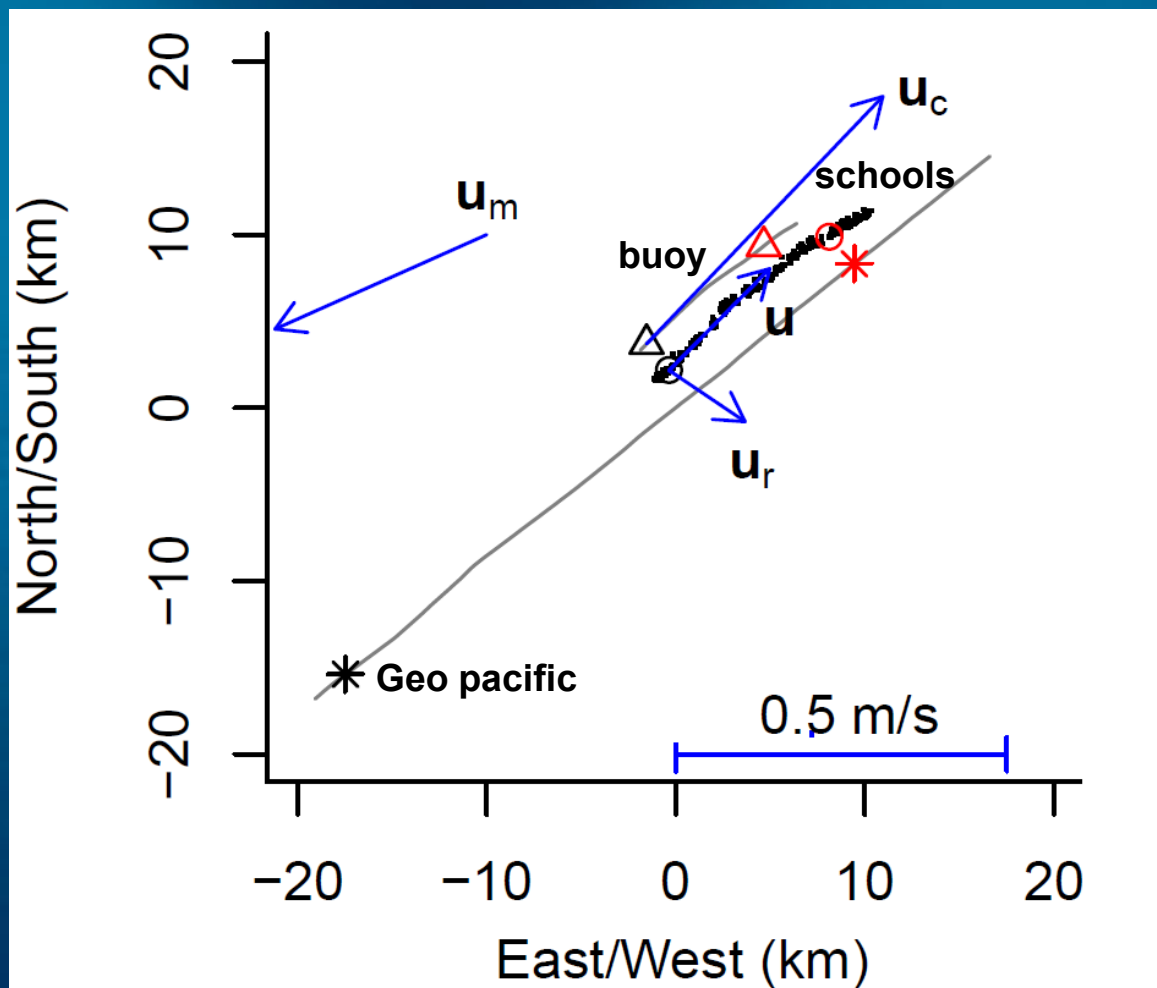
Protocols for 3D data acquisition

Pre-defined straight cruise line

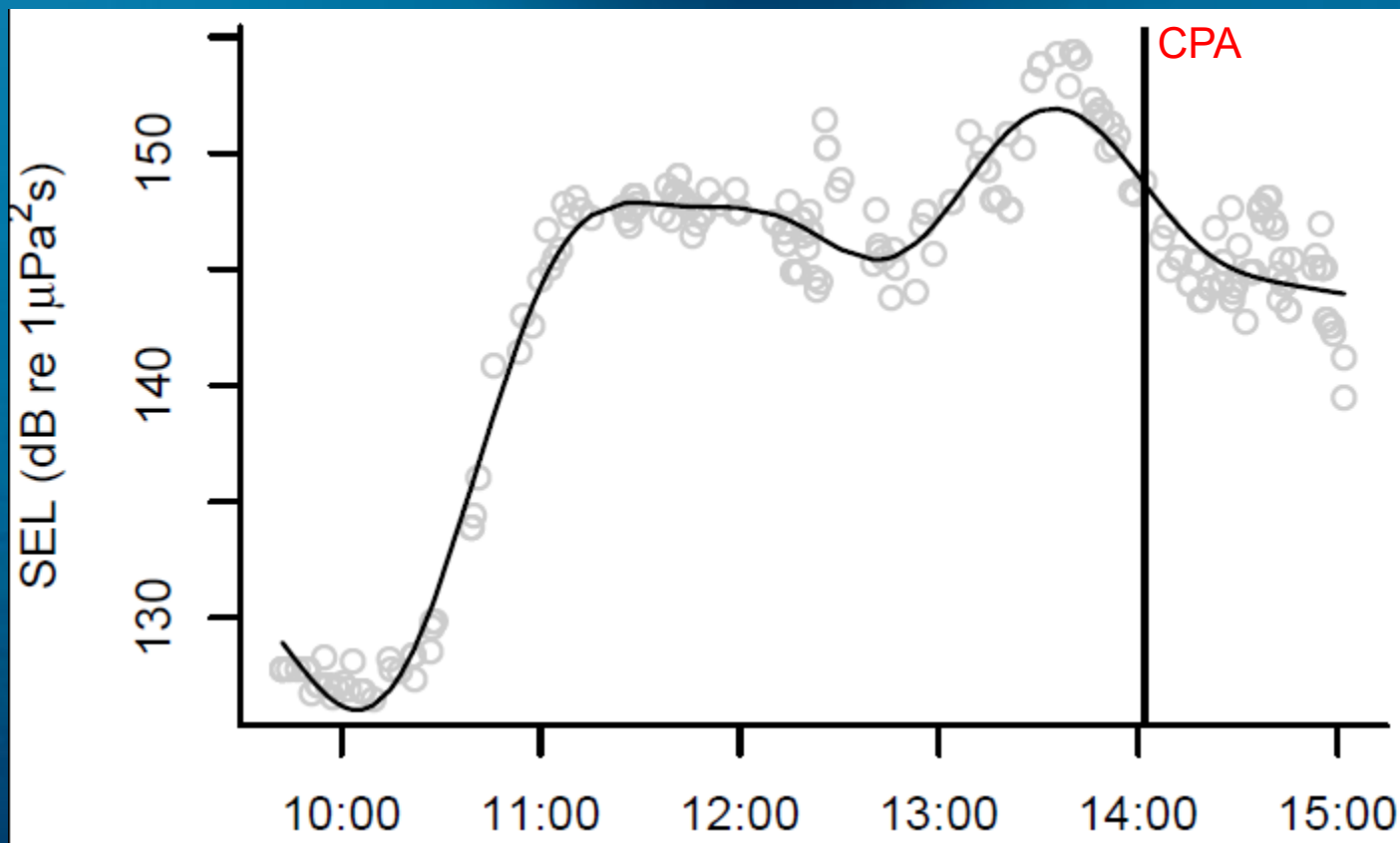
2 km minimum distance
between vessels



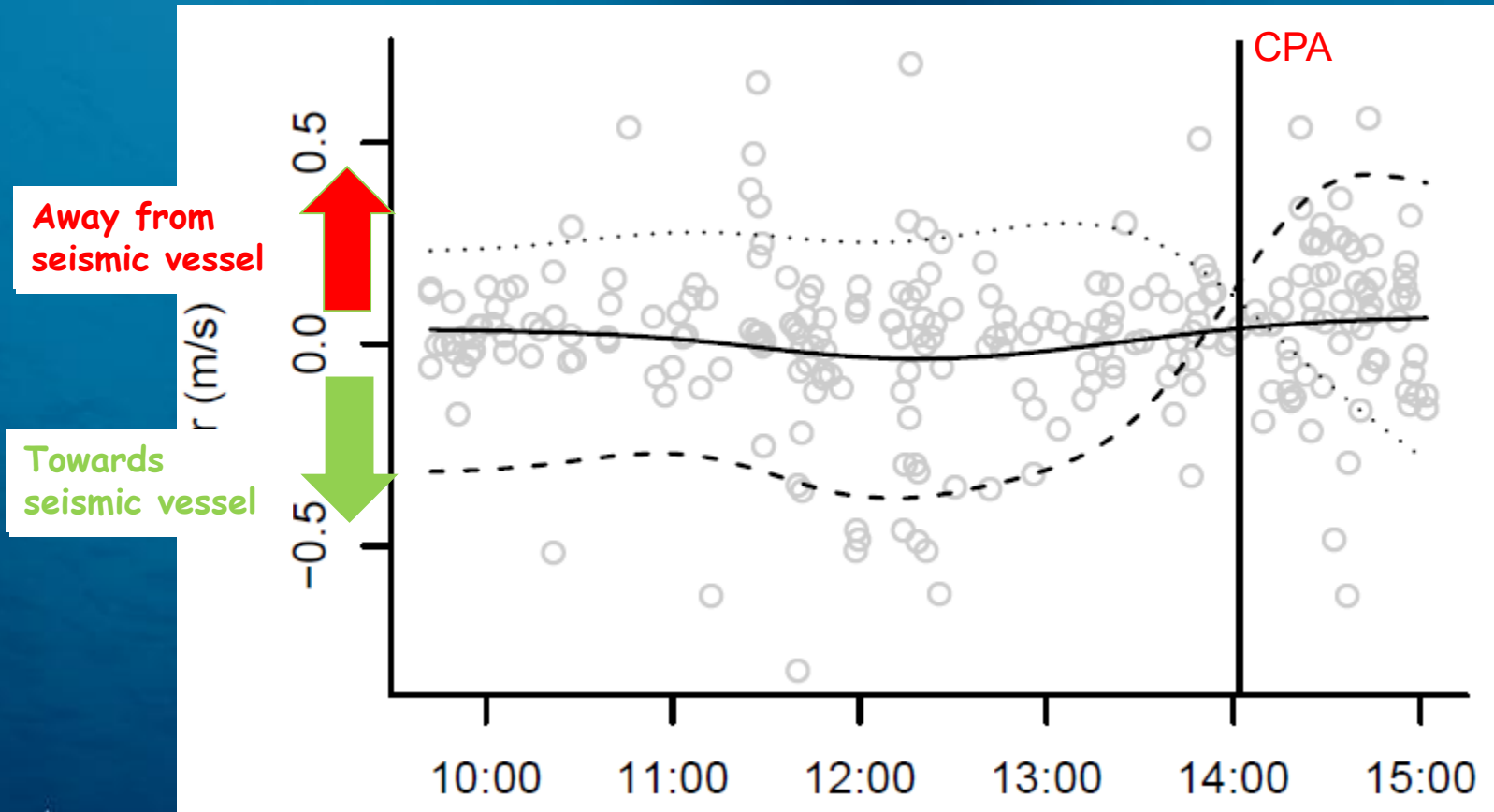
Overview of data and variables



Sound exposure level for each air gun



School directional response to seismic vessel



Investigation of bias in echosounder acoustic surveys using omnidirectional sonar from South to Northwest Africa

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⁵Institut Mauritanien de Recherches Océanographiques et des Pêches, **Mauritania**

⁶National Marine Information and Research Centre, **Namibia**

⁷National Institute for Fisheries and Marine Research, **Angola**

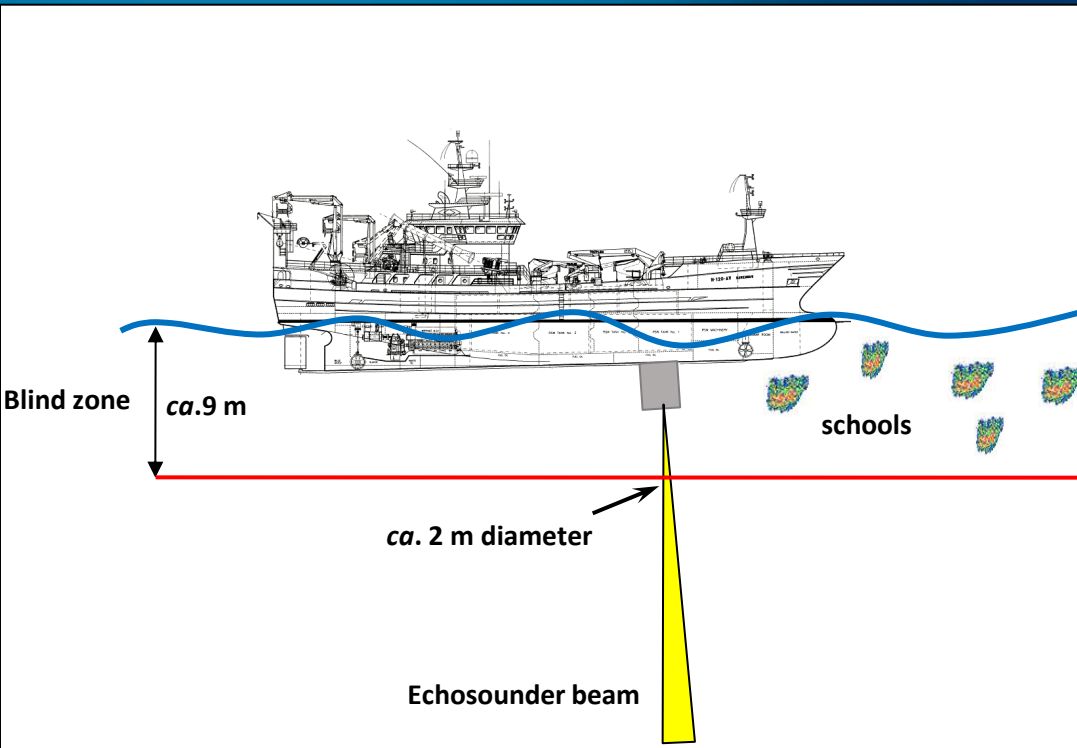


EAF-Nansen Programme

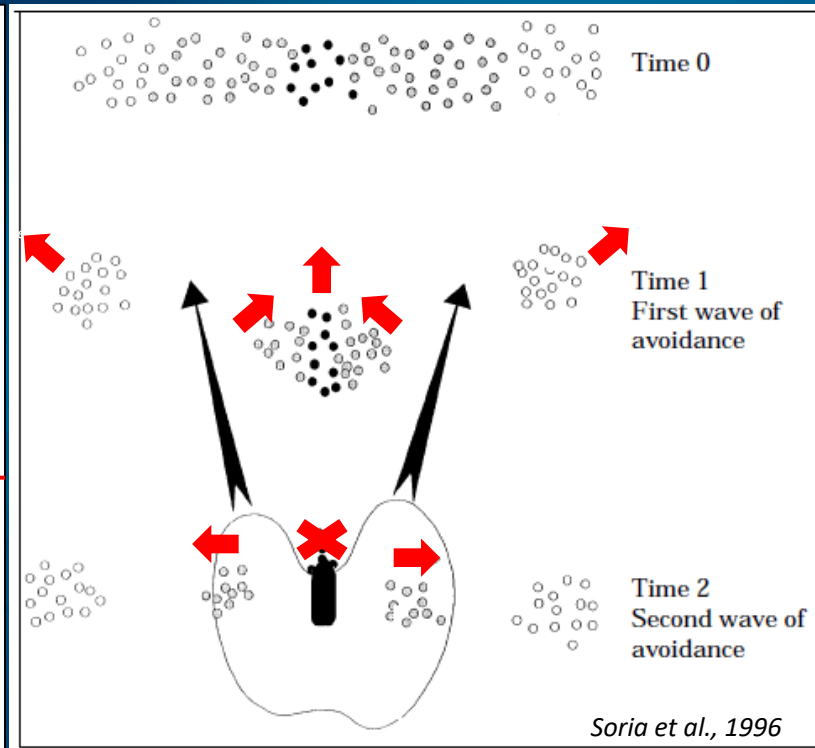


Background

Surface blind zone



Horizontal fish avoidance



Aim

Investigate possible bias in echo sounder measurements related with blind zone and fish avoidance



Post processing system LSSS used to handle echosounder and sonar data

10 to 30 m
Data used for
comparison with sonar
data

Echosounder

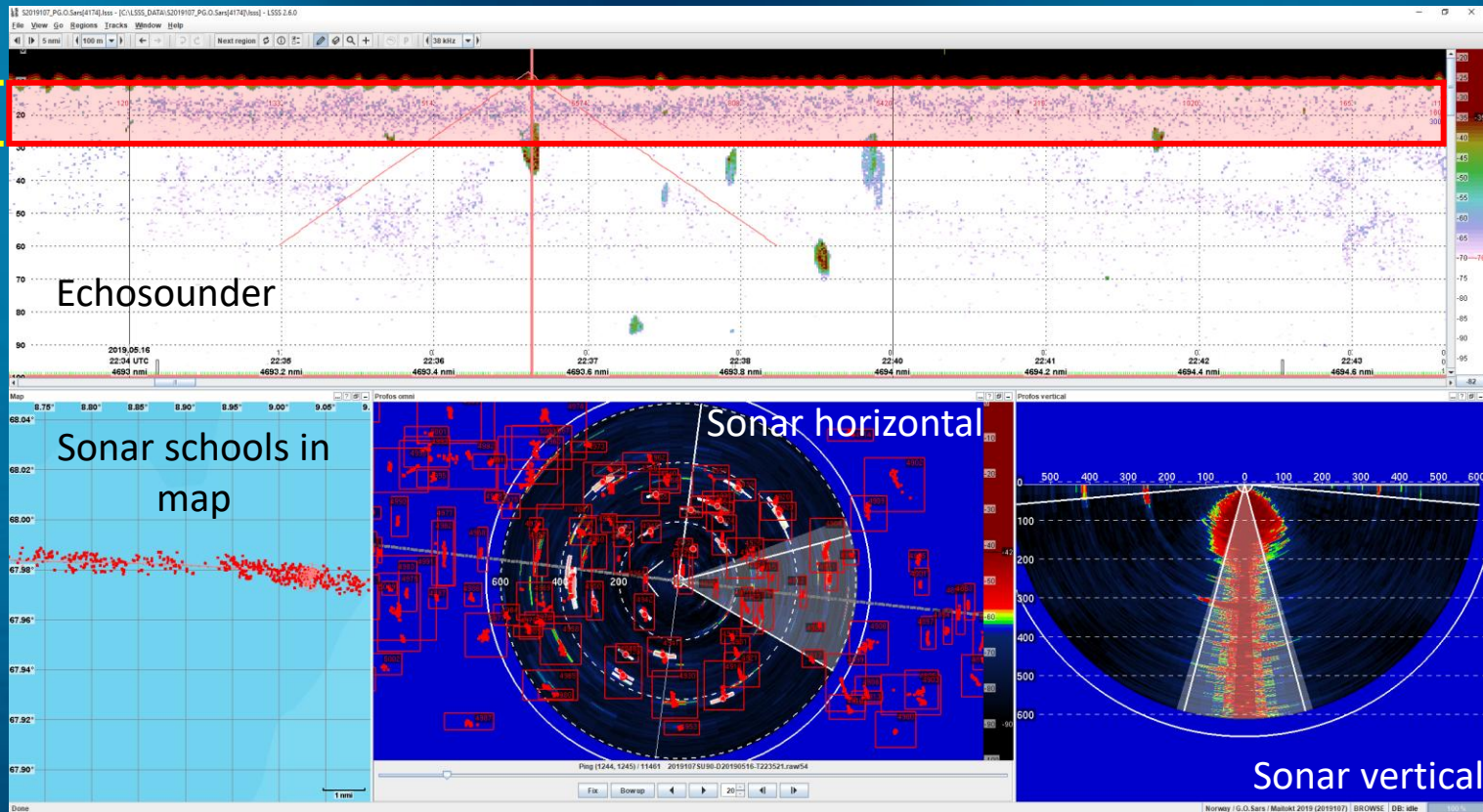
Sonar settings

360° horizontal
450 m range
3 deg tilt

Sonar schools in
map

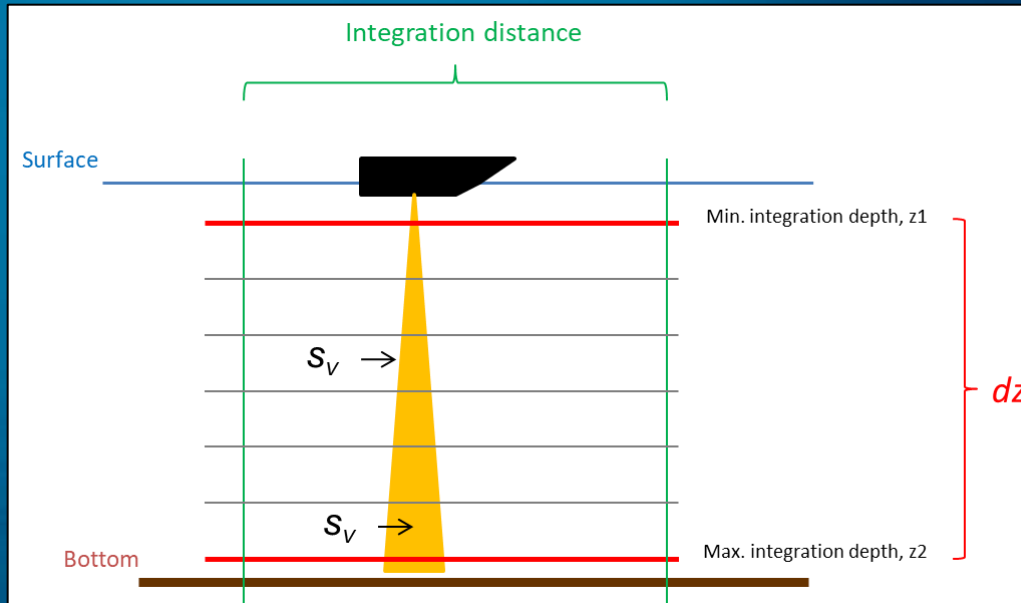
Sonar horizontal

Sonar vertical



Methods

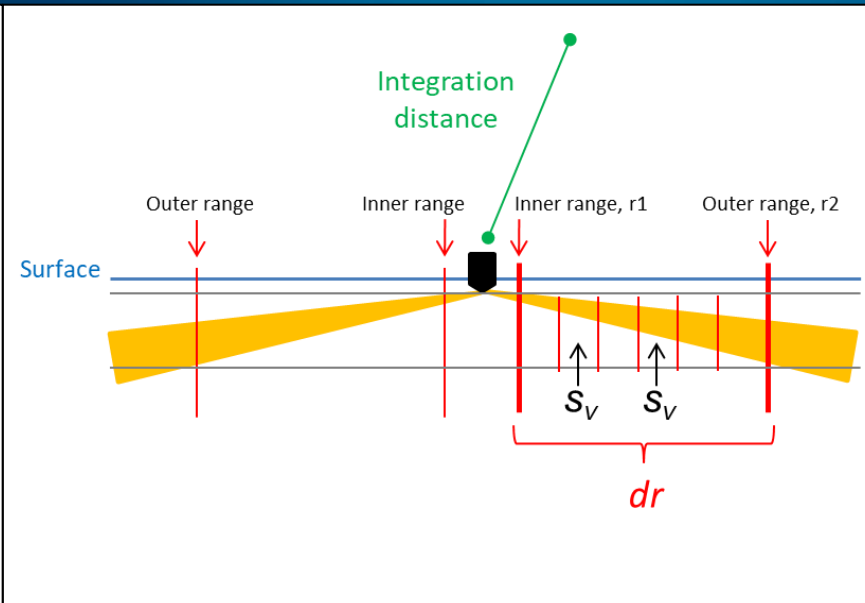
Echosounder **down-looking** echo integration



$$s_a = \int_{z1}^{z2} s_v dz$$

$$NASC = 4\pi(1852)^2 s_a$$

Sonar **side-looking** echo integration



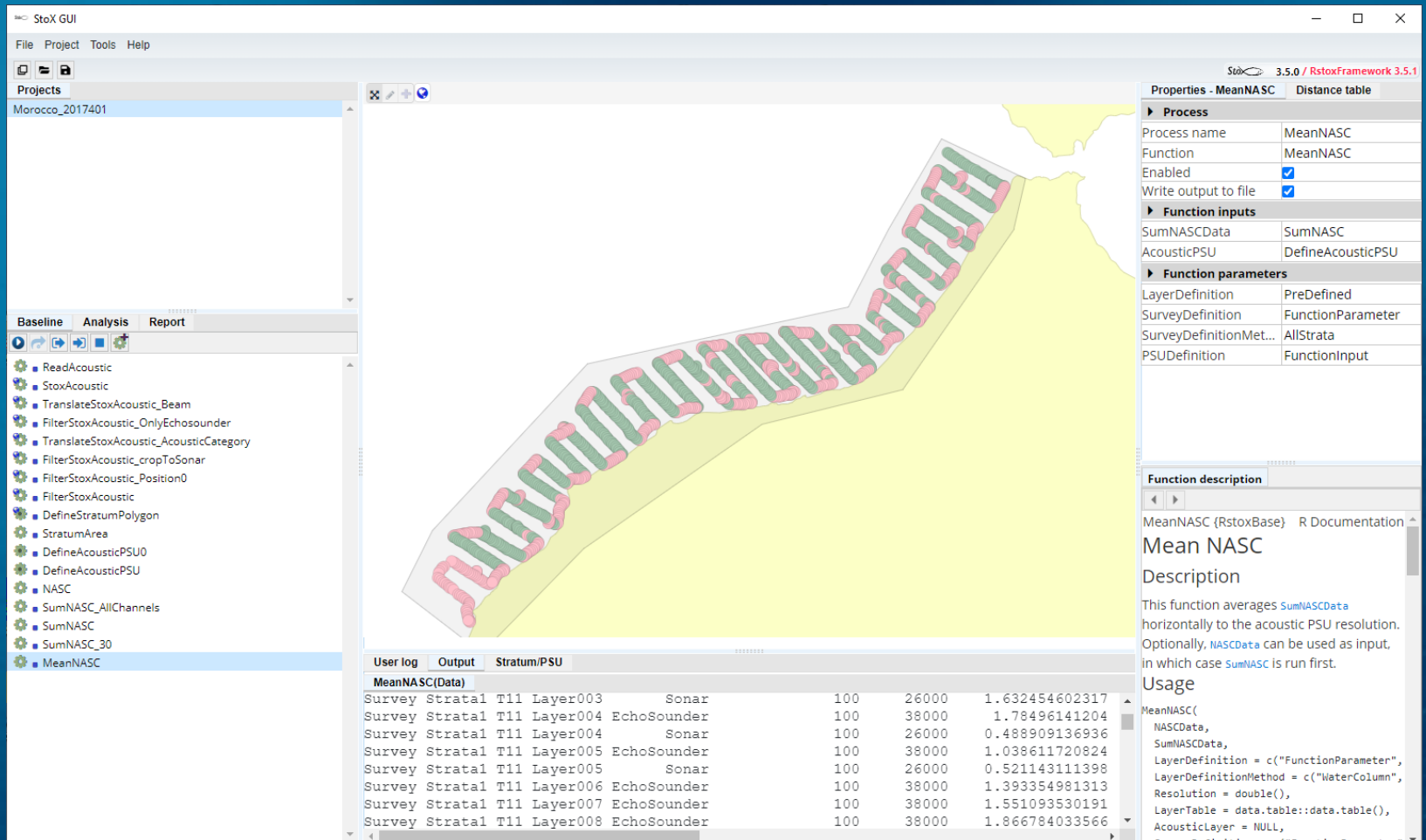
$$s_a = \int_{d1}^{d2} s_v dr \times 2.5$$

compensate
for lateral
aspect



Methods

StoX open-source software used for analyze sonar and echosounder acoustic NASC data

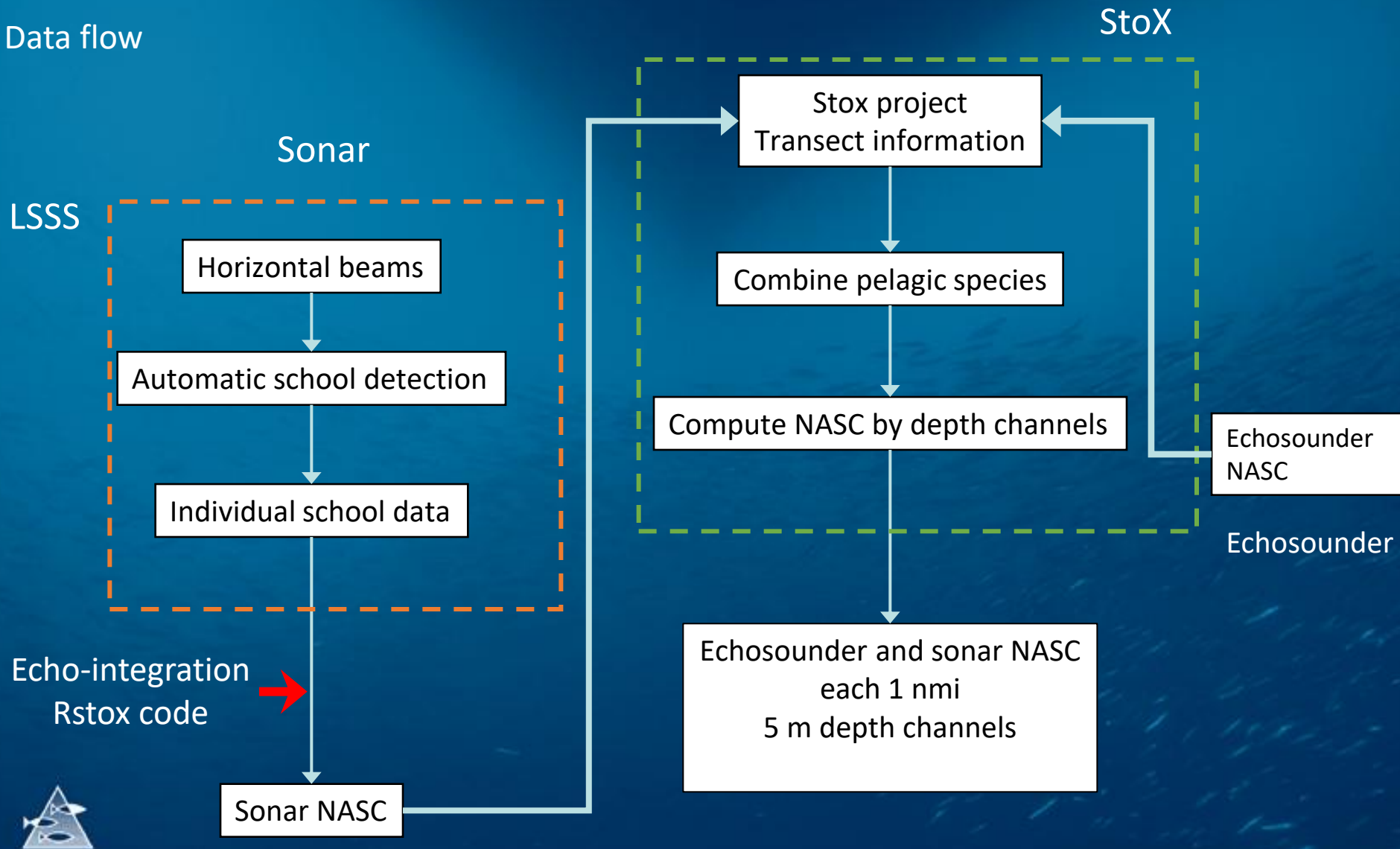


Johnsen E, Totland A, Skålevik Å, et al. StoX: An open source software for marine survey analyses. *Methods Ecol Evol.* 2019; 10:1523–1528.



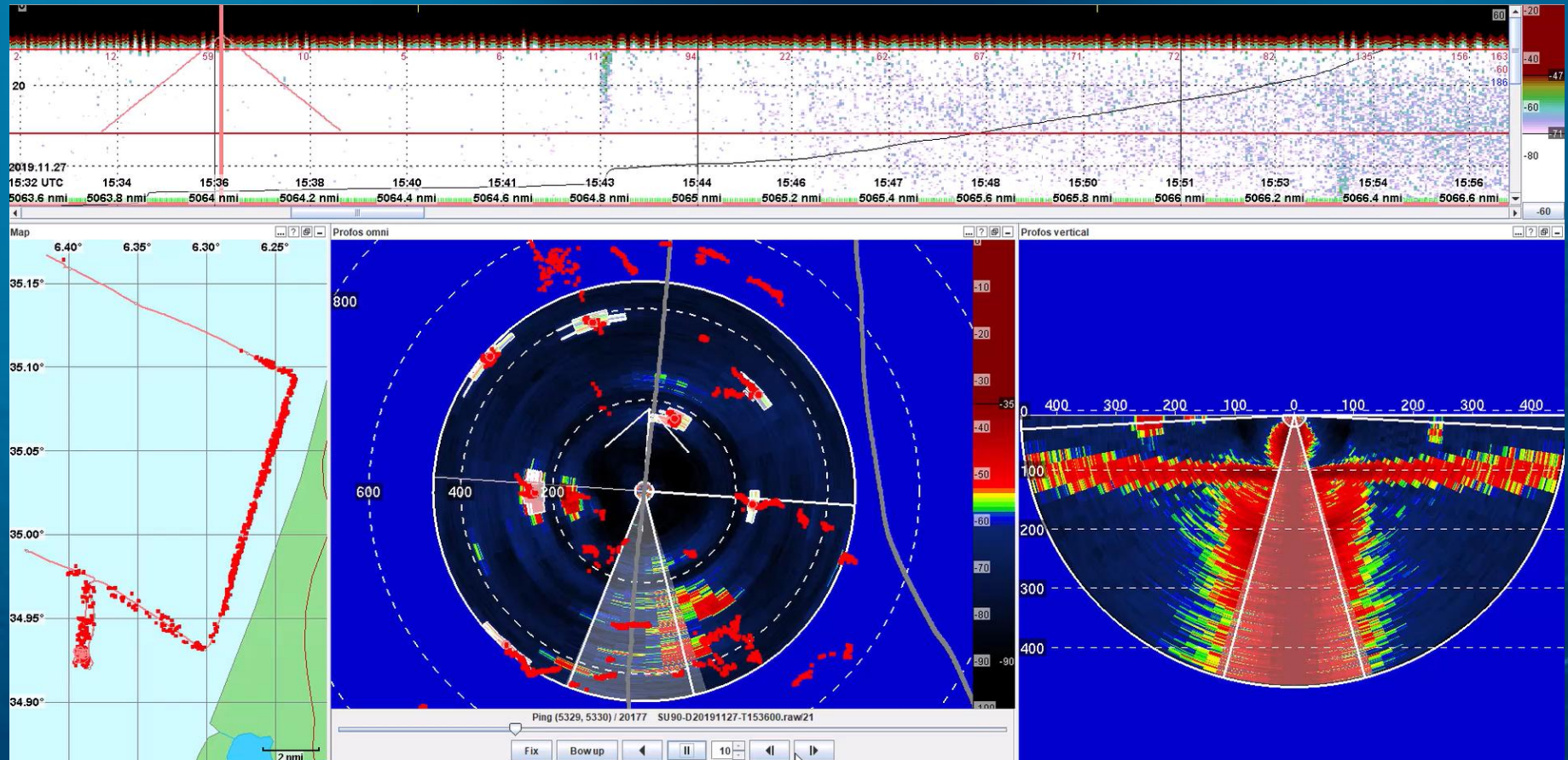
Methods

Data flow



Methods

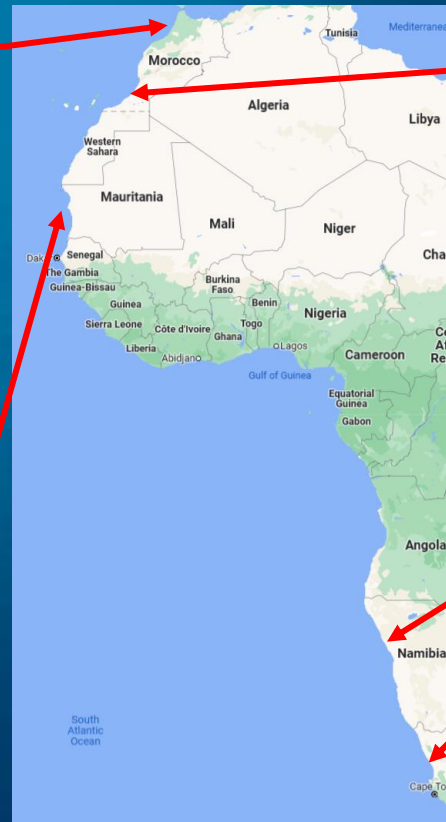
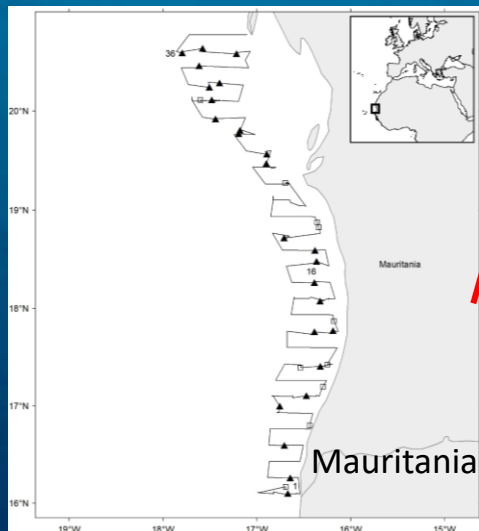
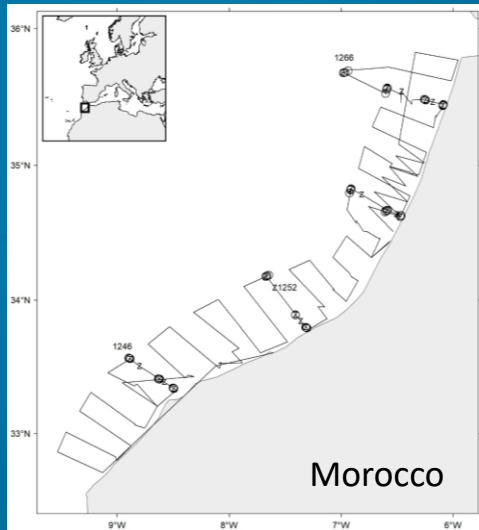
Example of echosounder and sonar data in Morocco 2019



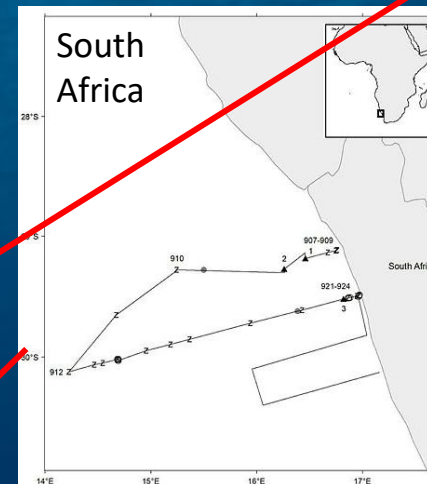
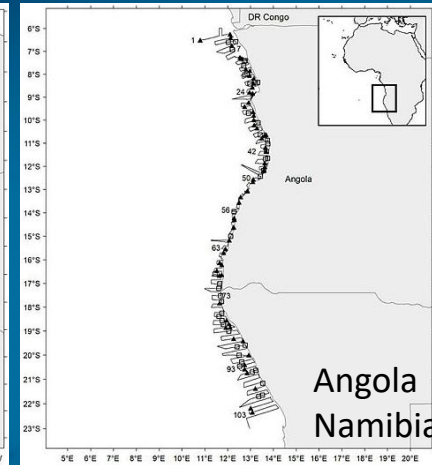
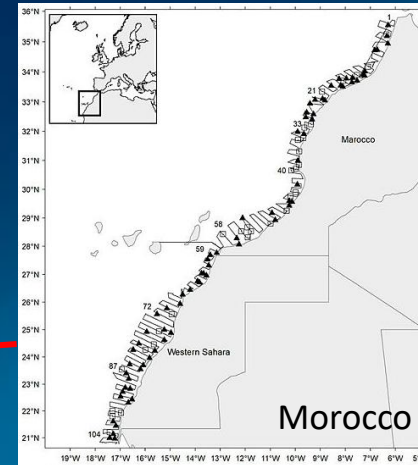
Acoustic data

R/V Dr. Fridtjof Nansen

2019



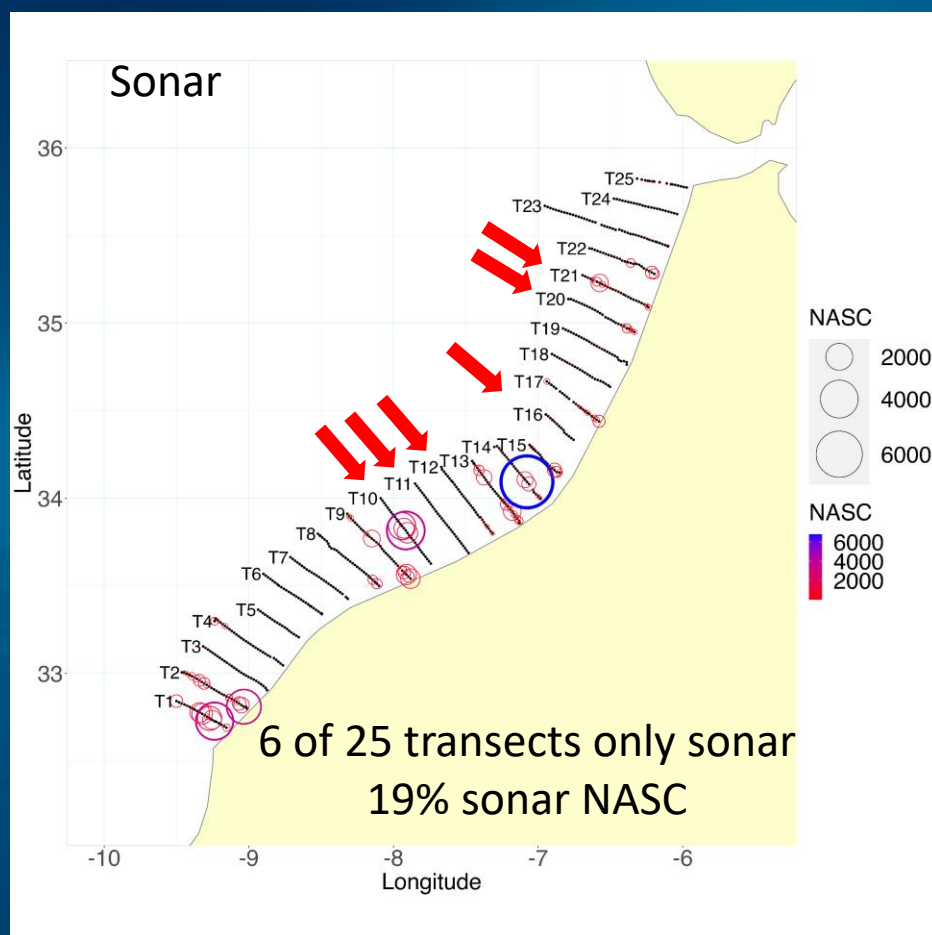
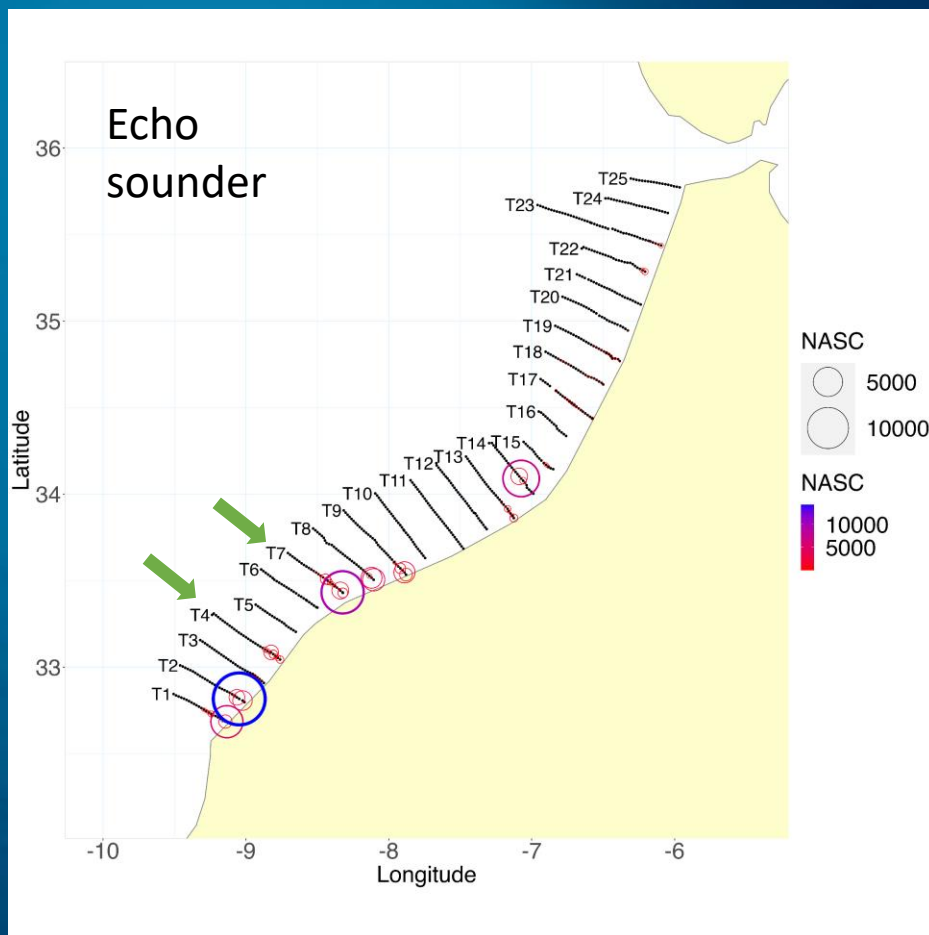
2017



Results

Morocco
2019

Abundance of pelagic fish in the upper 30 m

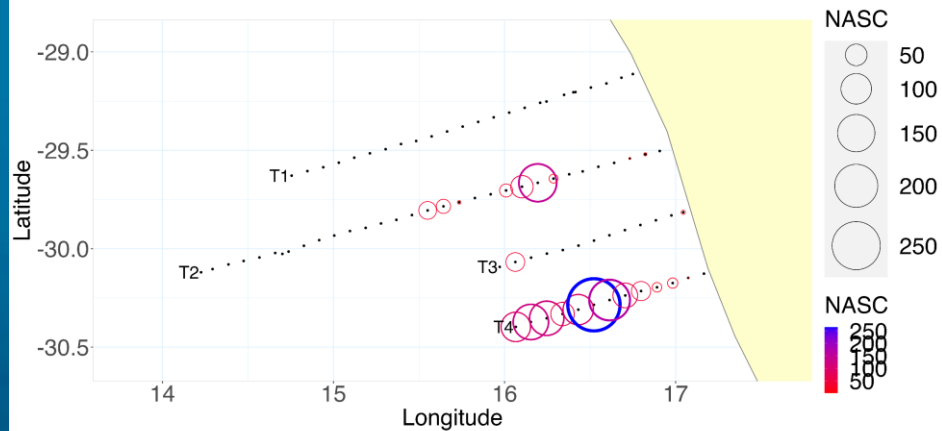


Results

South Africa
2019

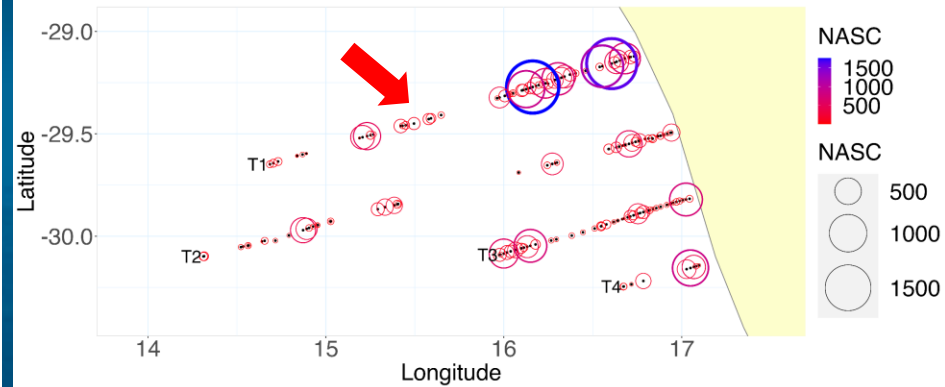
Abundance of pelagic fish in the upper 30 m

Echo sounder



Sonar

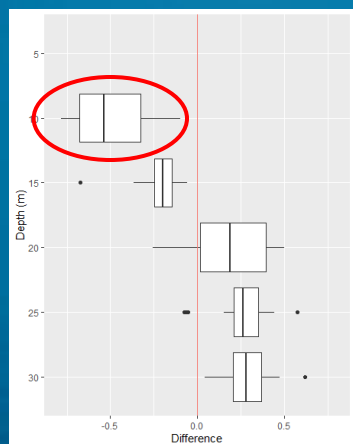
1 of 4 transects only sonar
37% sonar NASC



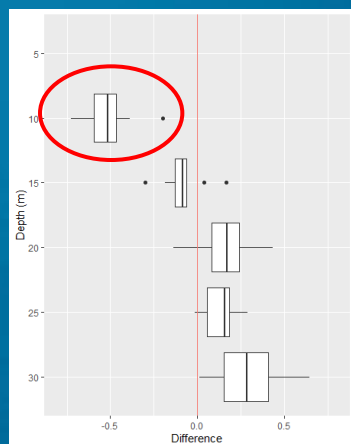
Results

Difference of NASC proportion between echosounder and sonar by depth channel

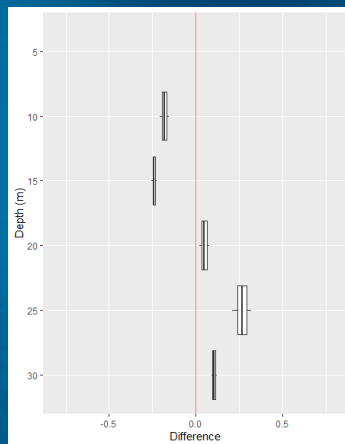
Morocco 2017



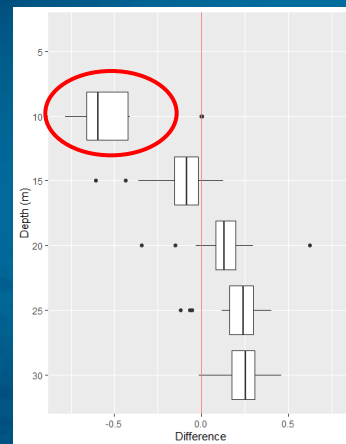
Namibia 2017



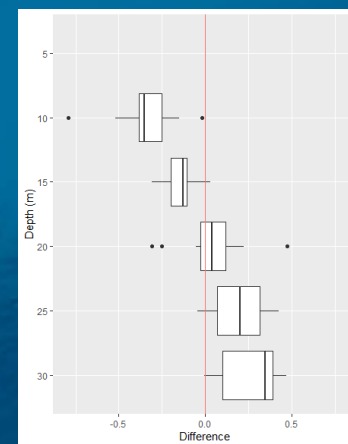
South Africa 2017*



Mauritania 2019



Morocco 2019*



High sonar High echo



Concluding remarks

- Sonar data collected and processed in 5 acoustic pelagic-trawl surveys in west Africa during 2017 and 2019
- **Procedure for sonar echo-integration and use of sonar data is proposed for routine surveys**
- Sonar not suitable to measure when pelagic fish is dispersing in layers (night hours)
- Transects with only sonar registrations account for 19 % (Morocco 2017) and 37% (South Africa 2017) of sonar NASC
- Data analyzed show a clear underestimation echo sounder measurements in Morocco and South Africa 2017
- Higher proportion of sonar estimates in all surveys in the upper 10 m relative to the echosounder
- The impact of the echosounder bias in the acoustic survey estimates needs to be evaluated

